Air Standards Cycle

① otto cycle → Petrol Engine → Spark Ignition Engine
② Diesel cycle → Diesel Engine → Compression Ignition Engine

★ Internal combustion engine :-

The combustion of fuel inside the cylinder in the presence of air take place and hard product of the combustion is directly act on the piston and do some external work.

→ Cooling → Air cooling (Bikes)
→ Water cooling (Cars)

→ Lubrication :- → Mist type
→ Splash Type

→ Bore :- Inside diameter of the cylinder.

→ T.D.C :- Top Dead Centre

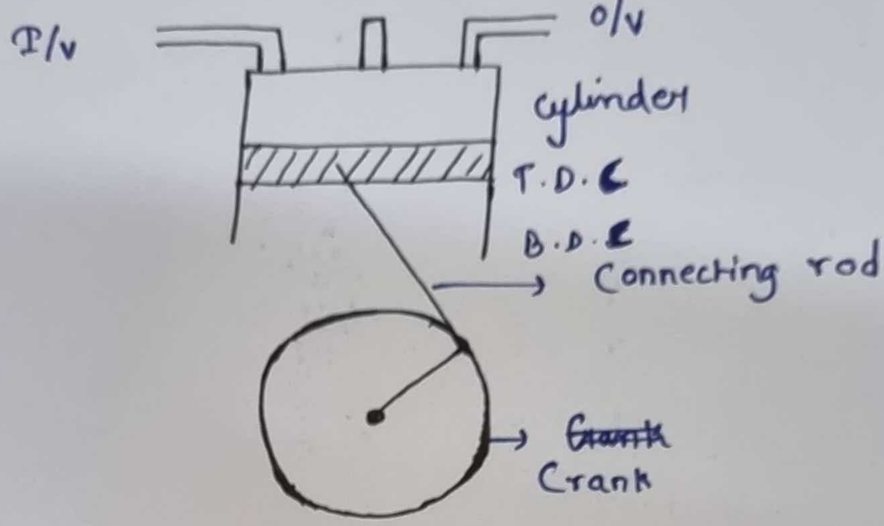
→ B.D.C → Bottom Dead centre

→ Stroke The distance travelled by piston between T.D.C and B.D.C

→ 4 - stroke :- less power than 2-stroke engine and also ~~consumber~~ consume less fuel than 2-stroke engine

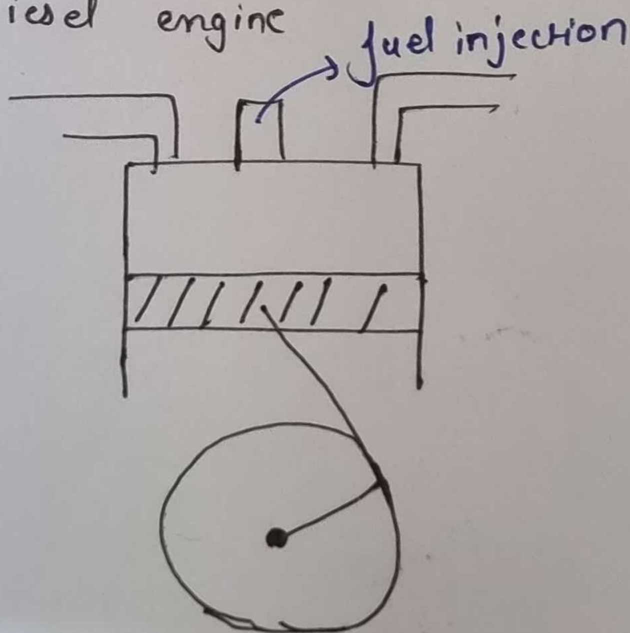
→ 2-stroke :-

4- Stroke petrol Engine:-

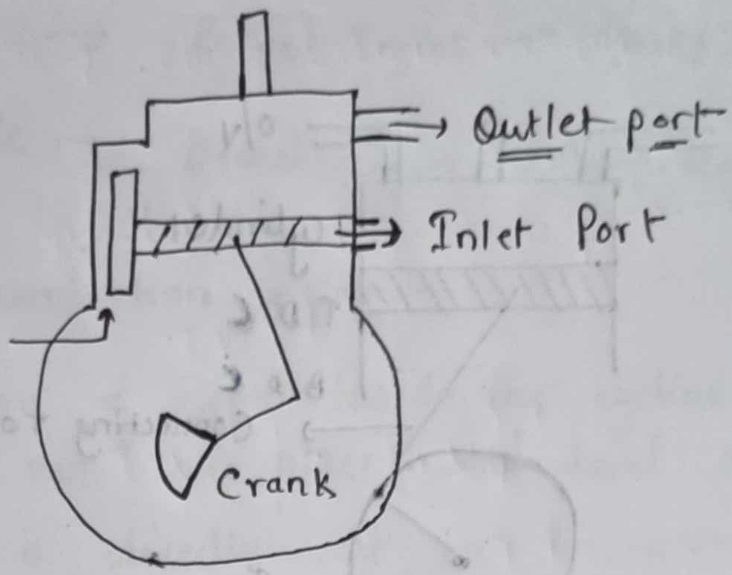


1. ~~Suct~~ Suction stroke :- TDC to B.D.C
2. Compression :- B.D.C to T.D.C
3. Expansion :- Power stroke.
4. Exhaust :-

→ ~~Disel~~ Diesel engine

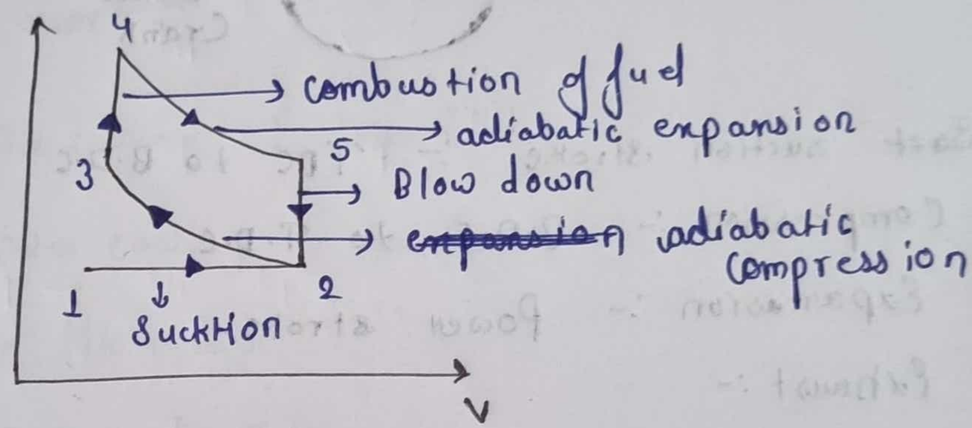


→ 2 Stroke petrol Engine.

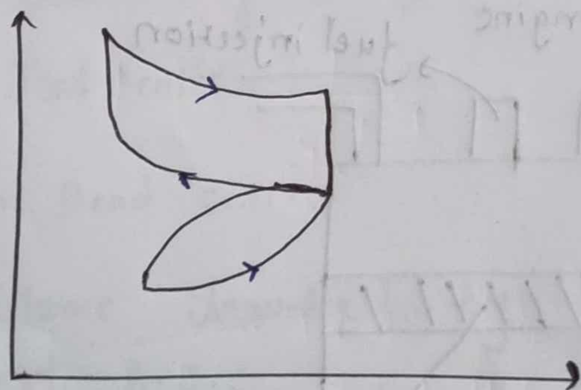


p-v diagram

Theoritical curve



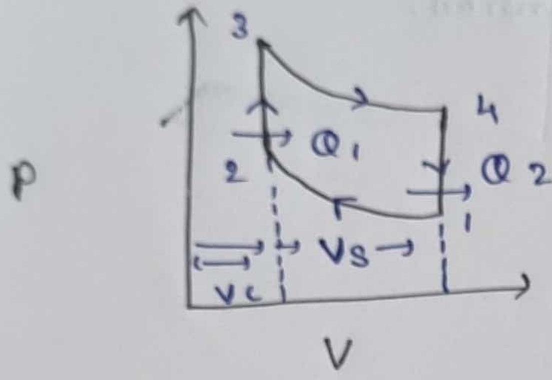
actual curve

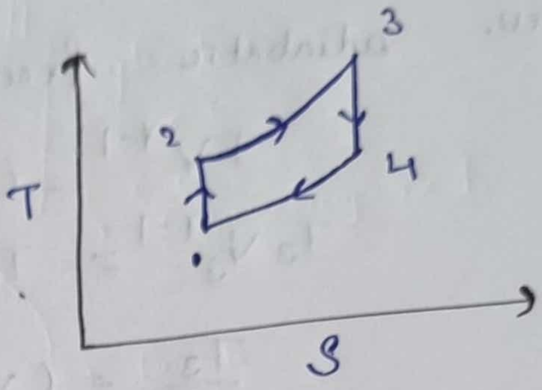


* Suction occur below ~~atmos~~ atmospheric pressure

The Valve should not opened and closed immediately.

<u>Otto cycle</u>





$$\eta = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1}$$

$$\Rightarrow 1 - \frac{Q_2}{Q_1}$$

$$\Rightarrow 1 - \frac{mC_v(T_4 - T_1)}{mC_v(T_3 - T_2)}$$

1-2 Rev. adiabatic process

$$TV^{\gamma-1} = C$$

$$T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$T_2 = T_1 \left(\frac{V_1}{V_2}\right)^{\gamma-1}$$

$$r_K = \frac{V_c + V_s}{V_c} = \frac{V_1}{V_2}$$

$V_c \to$ clearance Volume

$V_s \to$ Swept

$r_k \to$ compression ration

3-4 Rev. adiabatic Process expansion.

$$TV^{\gamma-1} = C$$

$$T_3 V_3^{\gamma-1} = T_4 V_4^{\gamma-1}$$

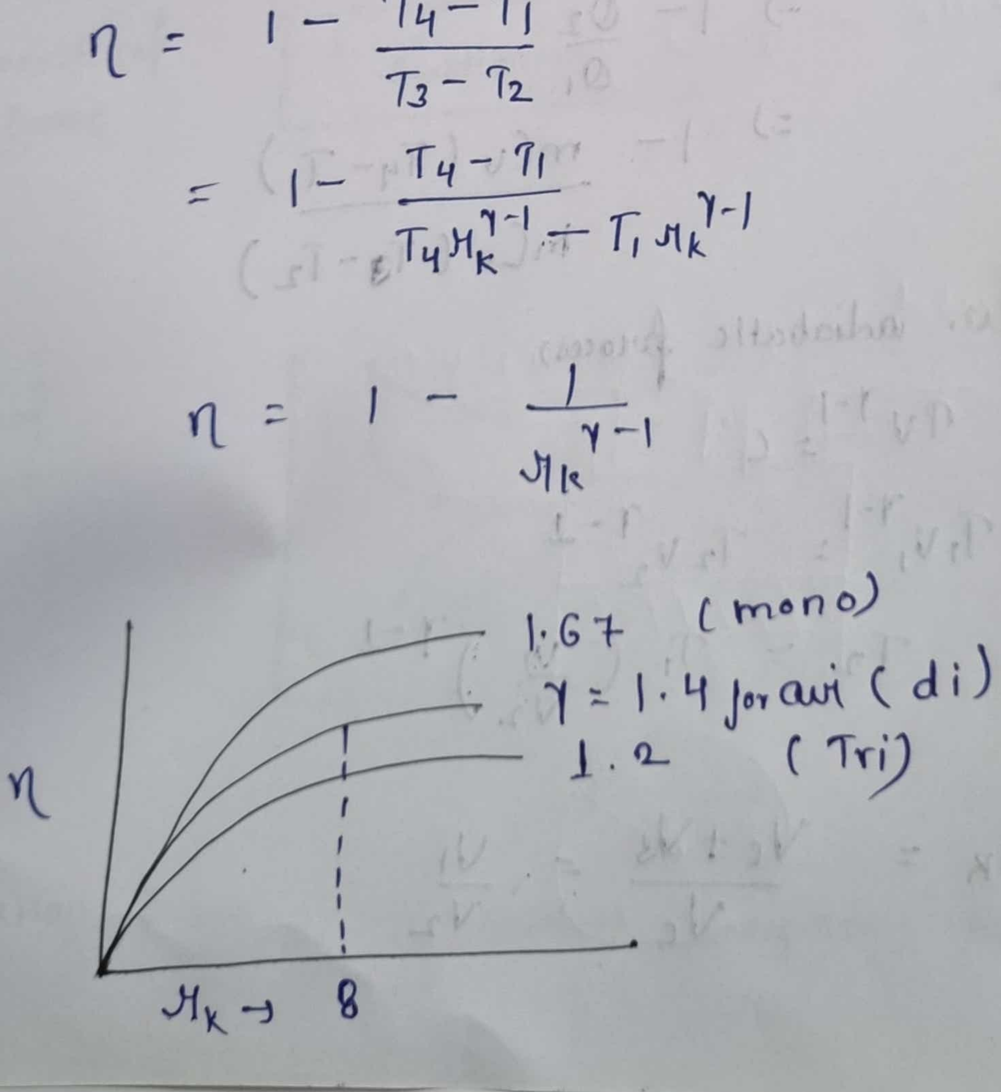
$$\frac{T_3}{T_4} = \left(\frac{V_4}{V_3}\right)^{\gamma-1}$$

$$T_3 = T_4 r_k^{\gamma-1}$$

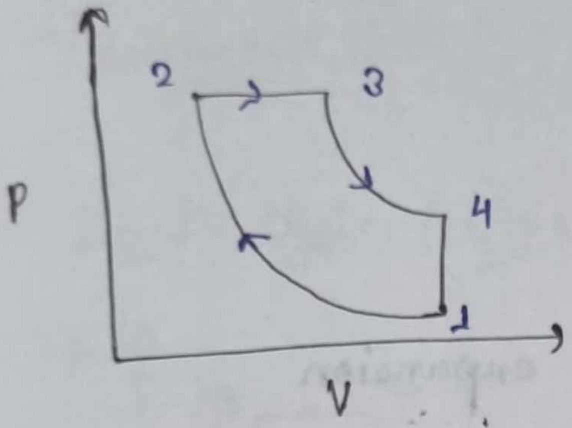
$$\eta = 1 - \frac{T_4 - T_1}{T_3 - T_2}$$

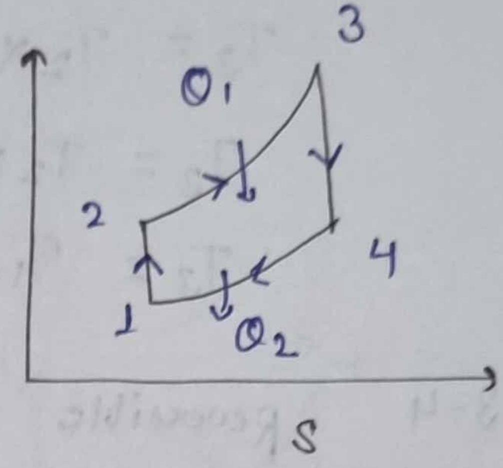
$$= 1 - \frac{T_4 - T_1}{T_4 r_k^{\gamma-1} - T_1 r_k^{\gamma-1}}$$

$$\eta = 1 - \frac{1}{r_k^{\gamma-1}}$$



Diesel Cycle





$$\eta = \frac{W}{Q_1}$$

$$= 1 - \frac{Q_2}{Q_1}$$

$$= 1 - \frac{\cancel{m}C_v(T_4 - T_1)}{\cancel{m}C_p(T_3 - T_2)}$$

$$= 1 - \frac{C_v(T_4 - T_1)}{C_p(T_3 - T_2)}$$

$$* \boxed{r_k = r_e r_c} *$$

$$\frac{V_1}{V_2} = \frac{\cancel{V_3}}{V_2} \times \frac{V_4}{\cancel{V_3}} = \frac{V_1}{V_2}$$

for 1-2 process is Reversible adiabatic process

$$T_2 = T_1 r_k^{\gamma - 1} \quad \text{— (1)}$$

for 2-3 process Reversible

$$\frac{V}{T} = C$$

$$\frac{V_3}{T_3} = \frac{V_2}{T_2}$$

$$T_3 = T_2 \times \left(\frac{V_3}{V_2}\right)$$

$$T_3 = T_2 \times r_c$$

$$T_3 = T_1 r_k^{\gamma-1} r_c$$

for 3-4 Reversible adiabatic expansion

$$T V^{\gamma-1} = C$$

$$T_3 V_3^{\gamma-1} = T_4 V_4^{\gamma-1}$$

$$\frac{T_3}{T_4} = \left(\frac{V_4}{V_3}\right)^{\gamma-1}$$

$$T_3 = T_4 r_e^{\gamma-1}$$

$$\frac{T_1 r_k^{\gamma-1} r_c}{\left(\frac{r_k}{r_c}\right)^{\gamma-1}} = T_4$$

$$T_4 = T_1 r_c^{\gamma}$$

$$\eta = 1 - \frac{1}{\gamma}\left[\frac{T_1 r_c^{\gamma} - T_1}{T_1 r_k^{\gamma-1} r_c - T_1 r_k^{\gamma-1}}\right]$$

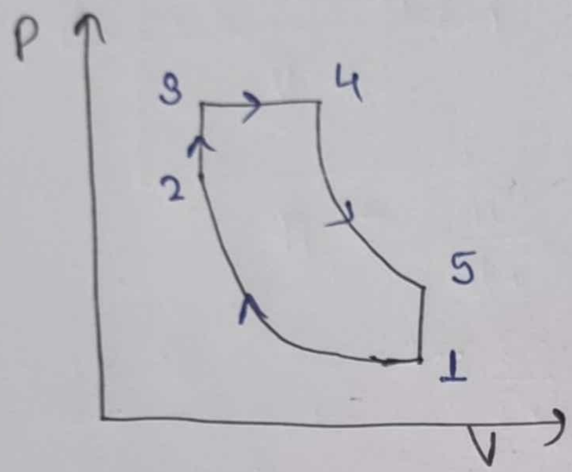
$$= 1 - \frac{1}{\gamma}\left[\frac{r_c^{\gamma} - 1}{r_k^{\gamma-1} r_c - r_k^{\gamma-1}}\right]$$

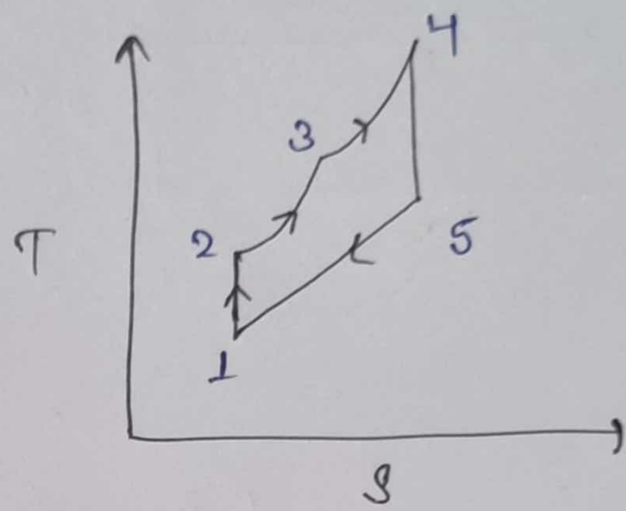
$$= 1 - \frac{1}{\gamma}\left[\frac{r_c^{\gamma} - 1}{r_k^{\gamma-1}(r_c - 1)}\right.$$

Cutoff (Important)

Ignition delay
continous flame

→ Dual cycle (viva)





→ Constant Volume and constant pressure Condition

NOTE :- * Diagram bana bhai paper me with pencil. *